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MULTI-SCALED MEASURES OF PAIN EXPERIENCE IN YOUNG ADULTS WITH SICKLE CELL DISEASE

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Daily experience of pain can lead to long-term disability, particularly for individuals with Sickle Cell Disease (SCD). Inheritance of this genetic disorder leads to changes in hemoglobin causing the stiffening and distorting of red blood cells into the shape of "sickles". This sickling blocks or partially restricts blood flow to different parts of their bodies, resulting in both chronic and acute pain. Current developments in mobile technology provide new tools for understanding and managing SCD pain outside the clinical setting, where characteristics of pain can be captured using self-report measures administered using mobile applications on patients' smart phones. However, self-report can be burdensome for patients and have potential iatrogenic effects. Mobile technology can capture characteristics of pain experience through the passive recording of physiological measures including movement and heart rate variability. Therefore, this project aimed to capture these passively recorded physiological signals across different reported pain intensities in order to identify measures that can be used as indicators of SCD pain experience. Four young-adult patients with SCD recruited from Cincinnati Children's Hospital Medical Center wore wristbands that recorded their heart rate, skin conductance, temperature, and movement (accelerometer data) for four consecutive days. They also used a mobile application to report their pain intensity, pain location, pain descriptors, happiness, stress and fatigue. The pattern-recognition techniques of machine learning were used to identify regularities in the physiological signals in relationship to the reported pain intensities. For example, statistical features from sliding windows of the accelerometer data were extracted to build personal models to predict the pain intensity for each patient. The Linear Regression, Multi-layer Perceptron and Random Forest classification algorithms significantly predicted pain intensity for two of the four patients. Future work will focus on the development of mobile health interventions that can leverage changes in physiological signals to provide adaptive support for pain management that is responsive to the momentary needs of individuals with SCD within their everyday lives.

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PRE-SURGICAL BIOPSYCHOSOCIAL CORRELATES OF LUMBAR FUSION OUTCOMES AMONG UTAH WORKERS' COMPENSATION PATIENTS

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Lumbar fusion surgery is an increasingly prevalent and costly procedure in the U.S. Outcomes following this surgery can be quite variable. There has been an effort to identify pre-surgical biopsychosocial variables that may predict lumbar fusion outcomes. Identification of such variables are particularly important for informing patient screening and selection procedures. The present purpose was to examine how pre-surgical demographic, health status, and compensation variables correlated with post-surgical outcome variables. Participants were 110 Utah workers' compensation patients (average age = 42.7 yrs., 77.3% male) who underwent lumbar fusion surgery from 1987–2007 and were at least two-years post-surgery at time of follow-up. A retrospective cohort design was utilized consisting of a review of pre-surgical records and a postsurgical telephone survey. Presurgical variables included: age at time of injury, number of levels fused, smoking status, amount of pain prior to surgery, depression history, assignment to a nurse case manager, and litigation status. The outcome measures were assessed at least 2-years post-surgery and included disability status, Roland-Morris Disability Questionnaire (RMDQ) total score, and Short-Form-36 Physical and Mental Health Composite Scores (PCS/MCS). Correlational analyses revealed that smoking, higher amount of pain prior to surgery, depression, assignment of a case manager, and litigation were all consistently statistically significant correlates of outcomes. Linear regressions were conducted in which pre-surgical variables were used to predict outcomes. Statistically significant amounts of variance were accounted for in disability status ($R^2 = 0.21$), RMDQ total ($R^2 = 0.26$), SF-36 MCS ($R^2 = 0.19$) and SF-36 PCS scores ($R^2 = 0.31$). The benefits of integrating pre-surgical predictors of outcomes into lumbar fusion patient selection protocols are discussed.

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